MILITARY SPECIFICATION

JACK, HYDRAULIC, TRIPOD MMU-45/E

1. SCOPE

1.1 This specification covers a 20-ton, hydraulic tripod jack, designated MMU-45/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed, and Lock Corner

Military

MIL-P-116	Preservation, Methods of
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-H-81019	Hydraulic Fluid, Petroleum Base, Ultra-Low Temperature

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods

DRAWING

Air Force

59J6185	Jack, Hydraulic, Tripod - MMU-45/E, 20 Ton Capacity, Assembly of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General requirements. The requirements of Drawing 59J6185 apply as requirements of this specification. When the drawing and this specification conflict, this specification shall govern.

3.3 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

3.4 Workmanship. The jack, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, and marking of parts and assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the jack shall be classified as follows:

a. Preproduction testing...See 4.4

b. Acceptance tests.....See 4.5.

4.3 Test conditions. The jack shall be serviced with hydraulic fluid conforming to MIL-H-5606 except for the low temperature test which shall be conducted with fluid conforming to MIL-H-81019.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One jack shall be subjected to the tests specified in 4.4.2.

4.4.2 Preproduction tests. The preproduction tests shall consist of all tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests

b. Sampling plan and tests.

4.5.1 Individual tests. Each jack shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Ram and cylinder hydrostatic pressure

c. Pump bypass relief valve test.

4.5.2 Sampling plan and tests

4.5.2.1 Periodic sampling. Five jacks shall be selected at random from every fifty or fraction thereof produced and subjected to the static proof test as described under 4.6.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items on hand or produced later shall be accepted until the event and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction all tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

investigation of a sampling test failure. But final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of this specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The jack shall be inspected to determine compliance with the requirements specified herein with respect to workmanship, marking, and compliance with the drawing.

4.6.2 Ram and cylinder hydrostatic pressure. The ram and cylinder assembly shall be proof-tested at 3,000 psi hydraulic pressure. Any leakage of hydraulic fluid shall be cause for rejection.

4.6.3 Pump bypass relief valve. The pump shall be proof-tested by sealing the pump outlet and operating the pump until the bypass valve opens. The bypass valve shall be adjusted as specified on the drawing.

4.6.4 Life test. With the screw extension retracted, the jack shall be subjected to the following tests:

4.6.4.1 Vertical lift. The vertical lift test shall consist of 200 cycles of raising and lowering a vertical load of 20 tons. One cycle shall consist of raising the hydraulic ram under load from the fully retracted position to the fully extended position and return to the fully retracted position. Failure of any part of the jack during this test shall be cause for rejection.

4.6.4.2 Dynamic side load. The dynamic side load test shall consist of 60 cycles of raising a 20-ton load vertically with a graduated 3-ton horizontal load. Twenty cycles each shall be applied through the jack pad socket adapter in the direction of an opposite jack leg. The horizontal load on the jack adapter shall be gradually increased from zero in the fully retracted position to 3 tons in the fully extended position. Failure of any part of the jack during this test shall be cause for rejection.

4.6.5 Static proof. After the bypass relief valve has been adjusted for overload, the jack shall be tested with the screw extension in the fully extended position by raising a 30-ton load vertically from the lowest position of the hydraulic ram to the fully extended position. While in the extended position, the vertical load shall be reduced to 20 tons and maintained during application of a 3-ton horizontal load applied to the top of the extended screw extension. To complete the cycle, the horizontal and vertical loads shall be removed and the jack ram allowed to retract to its original position with no more than 50 pounds applied to the top of the ram. Failure of any part of the jack during this test shall be cause for rejection.

4.6.6 Low temperature. The jack shall be subjected to low temperature in accordance with MIL-STD-810, method 502, procedure I. A temperature of -54°C shall be maintained for a period of 24 hours. At the end

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

of the exposure period and with the temperature maintained, the jack ram shall be raised to its maximum extension and pumped until the bypass relief valve operates. The lowering valve shall be opened, and the ram shall retract with not more than 50 pounds applied to the ram. For this test, the jack shall be serviced with hydraulic fluid conforming to MIL-H-81019. The time and number of full strokes necessary to fully extend the ram shall be recorded at both +21° and -54°C. The time and strokes required to fully extend the ram at -54°C shall be not more than double the time and strokes required at +21°C.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Cleaning. The cleaning process shall be in accordance with MIL-P-116, paragraph entitled C-1 Any applicable process.

5.1.1.1 Level A. Unless otherwise specified in the contract or order, each jack shall be preserved and packaged in accordance with MIL-P-116, method I. Each jack shall be in a disassembled condition with jack fully retracted. The tow bar shall be secured in an upright position to the jack leg.

5.1.1.2 Level C. The jack shall be preserved and packaged in accordance with the manufacturer's commercial practice. This level shall provide adequate protection against damage or deterioration during shipment from the supply source to the initial receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The jack preserved and packaged as indicated in 5.1.1.1 shall be packed in an export-type exterior container conforming to PPP-B-621. Strapping and closure shall be in accordance with the appendix to the applicable container specification.

5.2.2 Level B. The jack preserved and packaged as indicated in 5.1.1.1 shall be packed in a domestic-type exterior container conforming to PPP-B-621. Closure shall be in accordance with the appendix to the applicable container specification.

5.2.3 Level C. The jack preserved and packaged as specified in 5.1.1.2 shall be packed in accordance with the manufacturer's commercial practice. Packing shall be adequate to assure carrier acceptance and safe delivery at the lowest rate to the initial receiving activity. Containers shall conform to commercial freight handlers rules and regulations as applicable to the mode of transportation.

5.3 Physical protection. Interior blocking, bracing, cushioning, et cetera, shall be in accordance with MIL-STD-1186.

5.4 Marking. Unless otherwise specified in the contract or order, the interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Jack, Hydraulic, Tripod MMU-45/E.

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:33Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The MMU-45/E jack is intended for use as a wing jack for aircraft whose jack pads are between 52 and 197 inches from the ground.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Location and conditions for pre-production testing (see 4.4)

c. Selection of applicable level of preservation, packaging, and packing (see 5.1 and 5.2)

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Reviewers:
Air Force - 11, 82

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

SEG (SEPS)
WRIGHT-PATTERSON AFB, OHIO

UNITED STATES AIR FORCE
OFFICIAL BUSINESS

Commander
Systems Engineering Group
ATTN: SEPS
Wright-Patterson AFB, Ohio

FOLD

PLATE NO. 15419 (BACK)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.